EcoLoop Student Setup Guide

Agentic coding, free local AI models, and zero-cost web deployment.

Swami Sant Dass Public School
github.com/Harman209/ecoloop

🎓 MIT Missing Semester • Lecture 7: Agentic Coding

MUST WATCH

Taught at **MIT CSAIL** (missing.csail.mit.edu). This lecture bridges the gap between basic chat tools and actual autonomous development. It covers tool use (how agents read and write real workspace files), execution loops (letting agents run compilers and tests to self-repair), and context management via `AGENTS.md`.

Search on YouTube: **"MIT Missing Semester Lecture 7: Agentic Coding"** • Link: **missing.csail.mit.edu**

1. The Paradigm Shift: Developer as Architect

CORE CONCEPT

Traditional coding required months of syntax memorization before you could test an idea. With modern agentic harnesses, you don't type boilerplate by hand. You define the rules, architecture, and user experience; an autonomous agent creates the components and updates repository files; your job is quality control and browser testing.

2. Quick Tool Setup: OpenCode Terminal Agent

60 SECONDS

OpenCode (github.com/anomalyco/opencode) is an open-source terminal coding agent that runs on Linux, macOS, and Windows (WSL/PowerShell). It reads your local project tree and executes multi-file edits directly.

```
# Option A: Install globally via npm (Node.js)
npm install -g @opencode/cli

# Option B: Direct shell installer (macOS / Linux)
curl -fsSL https://opencode.ai/install.sh | bash
```

3. The Free AI Matrix: Zero Dollars, No Credit Cards

STUDENT FRIENDLY

You do not need expensive subscriptions to run autonomous coding agents. Here are the 4 best zero-dollar options:

1. Ollama • 100% Offline & Private

Download **ollama.com**. Runs entirely on your laptop hardware without internet. Run `ollama run qwen2.5-coder:7b` for top-tier coding performance on standard student laptops.

2. Google AI Studio (Gemini API)

Visit **aistudio.google.com**. Generates free API keys with generous rate limits for Gemini 1.5 Flash and Gemini 2.0. Perfect for large full-stack codebases.

3. Groq Cloud • Ultra-Fast Inference

Visit **console.groq.com**. Provides ultra-fast token streaming of open-weight models like Llama 3.3 70B for $0.00. Instant file generation.

4. GitHub Student Developer Pack

Verify your student ID at **education.github.com**. Unlocks free access to GitHub Models, Codespaces, domain names, and free cloud hosting credits.

Masterclass Library & Project Deployment

Curated video tutorials and step-by-step instructions to launch your own circular platform.

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ecoloop-ssdps.vercel.app

4. Curated Video Masterclasses RECOMMENDED WATCHING

1. MIT CSAIL: The Missing Semester • Lecture 7: Agentic Coding

`YouTube Search: "MIT Missing Semester Lecture 7 Agentic Coding"`

Understand the underlying mechanics: tool execution loops, context boundaries, file system access, and how agents self-diagnose compiler errors.

2. Ollama • Complete Beginner Guide to Running AI Locally on Your Laptop

`YouTube Search: "Ollama beginner guide local AI models tutorial"`

How to install Ollama on Windows, Mac, or Linux, pull `qwen2.5-coder`, and connect it to your terminal tools with zero cloud reliance.

3. Deploy React & Vite to Vercel in 60 Seconds (100% Free Hobby Tier)

`YouTube Search: "Deploy Vite React app to Vercel free guide"`

Connect your GitHub repository to Vercel for automatic continuous deployments, custom domain routing, and free global SSL.

4. Google Sheets as a Free Database with Google Apps Script Webhooks

`YouTube Search: "Google Sheets as database Google Apps Script POST webhook"`

How to write a 10-line Apps Script function that receives POST requests from a React form and appends new rows into a spreadsheet for $0/month.

5. How to Fork & Deploy EcoLoop for Your School TAKE ACTION

1. **Fork the Repository:** Go to `github.com/Harman209/ecoloop` and click *Fork* to create your own copy on GitHub.
2. **Deploy on Vercel:** Sign in to `vercel.com` with GitHub, click *Add New Project*, select your forked `ecoloop` repo, and click *Deploy*. You get a live `.vercel.app` URL in under 2 minutes.
3. **Connect Your Google Sheet:** Create a Google Sheet, open *Extensions → Apps Script*, paste the webhook script from `docs/apps-script.js`, and paste your web app URL into your project's contact form. All inquiries now flow into your private sheet.

6. The Three Invariant Rules of Working with Coding Agents BUILDER WISDOM

Rule 01: Write the Spec First (AGENTS.md)

Create an `AGENTS.md` file in your repo root before coding. State your colors, design rules, and file boundaries so the agent never hallucinates.

Rule 02: Deployed Without Testing = Broken

The AI will confidently say "all bugs resolved". Never believe it. Click every button, test the form fields, and inspect the browser console yourself.